

Rahul Guruprakash

rahulguruprakash@gmail.com | +1 (858) 305-6369 | [g-rahul.github.io](https://github.com/rahul-g) | linkedin.com/in/rahul-g-055720175

Education

North Carolina State University

Master of Science in Computer Engineering

Relevant Coursework: Advanced CPU Microarchitecture, Microarchitecture Security, Operating Systems
Parallel Computing Architecture, General-Purpose GPU Architecture

Raleigh, NC

GPA: 4.0/4.0

R.V. College of Engineering

Bachelor of Electrical and Electronics Engineering

Bangalore, India

GPA: 8.3/10

Skills

Programming Languages: Embedded C/C++, AArch64 assembly, Python, Shell, Perl.

Build Systems: CMake, SCons, GNU/LLVM, Buildroot, BusyBox, Yocto.

Debug Tools: Trace32, GDB, OpenOCD, QEMU, Arm FVP, GTest.

System Software: Trusted Firmware-A, Hafnium, Xen, Linux kernel drivers, MCUBoot, NuttX, FreeRTOS.

Hardware Design: Altium, KiCad.

Professional Work Experience

Arm - Engineer, System Compliance - ATG, Bangalore

July 2023 – August 2025

- Maintained and contributed to Arm's open-source [FF-A Architecture Compliance Suite \(ACS\)](#) for A-profile platform compliance, writing portable C and assembly test cases.
- Designed FF-A v1.1 and v1.2 compliance tests for secure partition memory and interrupt models, runtime state transitions, and Secure and Normal world communication paths.
- Built ACS platform support for Arm Base FVP targets, adding AArch64 bring-up code and MMU, GIC, SMMU, and DMA validation flows.
- Designed an ACS Linux kernel driver for user-space FF-A compliance and added NS-Hypervisor support for Android pKVM validation.
- Contributed FF-A fixes to reference implementations, including Hafnium (SPMC), Xen (NS-Hypervisor), and TF-A.

Qualcomm - Engineer / Associate Engineer, RF-Link Manager Team, Hyderabad

October 2021 – July 2023

- Contributed to Snapdragon X35 5G modem RF bring-up on the RF command/control path, implementing firmware support for script execution, transceiver boot, front-end programming, and RF configuration across RAT modes.
- Built and maintained GTest-based validation for pre-silicon RF subsystem bring-up, using virtual platforms to surface firmware, RTL, and early-silicon issues.
- Designed CRC checks for AXI linked-list DMA programming traffic between the modem DSP and RF subsystem to detect RF front-end programming corruption.
- Introduced section-gapped RF firmware images to work around an instruction-decoder hardware issue, coordinating compiler support with the tools team.

Log9 Materials - Engineer, Embedded Systems, Bangalore

January 2021 – September 2021

- Built charging-controller firmware for Electric Vehicle Charging Controller (EVCC) hardware used in three-wheeled electric autos.
- Developed the EV charging protocol stack with an OSI-layered implementation of GB/T 27930 and SAE J1939/21 over CAN, adding diagnostics and in-field integration support.
- Designed EVCC front-end hardware in Altium around a low-power Cortex-M4 SoC, integrating CAN, power, memory, and battery management interfaces.
- Brought EVCC hardware into production deployment across 100+ three-wheeled electric autos, including battery management system integration.

Academic Research

Graduate Research Assistant - North Carolina State University

August 2025 – Present

- Develop embedded AI hardware platforms for biomass mapping and precision agriculture workflows.
- Build a custom carrier board for the NVIDIA Orin NX compute module, covering signal integrity and high-speed layout for PCIe Gen4, GbE, and USB 3.2.